

Make decisions under uncertainty

A decision you can explain

Connect the four branches below

1. Name the reference group

Conditional probability restricts the denominator to the condition. $P(A | B)$ need not equal $P(B | A)$. Base rates matter.

2. Choose across uncertain states

Expected value weights outcomes by probability. Probabilities sum to one, and payoffs use compatible units and horizons.

3. Model and simulate

Simulation explores assumptions. Its output is an estimate, not population truth. Inspect reproducibility and trial counts.

4. Communicate bounded risk

Explain which assumption could reverse the action. A fifth percentile is a cut-off with roughly 5% below it, not a guaranteed minimum.

Name the reference group

Event A and $B > P(A | B)$

The connection

Conditional probability restricts the denominator to the condition. $P(A | B)$ need not equal $P(B | A)$. Base rates matter.

A business example

A screen can detect most true cases while many positive screens are false alarms when the event is rare.

Explain it without looking

Name the observational unit, the evidence needed, and one limitation. Connect this branch to the next decision in the session.

Choose across uncertain states

States > probabilities > payoffs > expected value

The connection

Expected value weights outcomes by probability. Probabilities sum to one, and payoffs use compatible units and horizons.

A business example

Higher expected profit may come with a larger adverse-state loss. Compare both.

Explain it without looking

Name the observational unit, the evidence needed, and one limitation. Connect this branch to the next decision in the session.

Model and simulate

Inputs > trial logic > outputs > sensitivity

The connection

Simulation explores assumptions. Its output is an estimate, not population truth. Inspect reproducibility and trial counts.

A business example

Separate an analytic expectation from a particular run's mean. Check units, boundary inputs, and downside.

Explain it without looking

Name the observational unit, the evidence needed, and one limitation. Connect this branch to the next decision in the session.

Communicate bounded risk

Expected result + downside + reversal condition

The connection

Explain which assumption could reverse the action. A fifth percentile is a cut-off with roughly 5% below it, not a guaranteed minimum.

A business example

Name the threshold, loss risk, and next evidence before committing resources.

Explain it without looking

Name the observational unit, the evidence needed, and one limitation. Connect this branch to the next decision in the session.